Introduction

The log-normal survival model assumes that $\log T$ follows a Normal distribution. Its distinctive feature among parametric survival models is a *non-monotone* hazard — the instantaneous risk rises early in follow-up, peaks, and then declines. This shape suits diseases or processes in which susceptible individuals fail relatively quickly while a hardy subset survives indefinitely: for example, post-surgical mortality (early peri-operative risk that wanes), some cancer recurrences, and time-to-engagement processes in behavioural data. Where Weibull and exponential models lock in monotone hazards, log-normal accommodates the rise-and-fall pattern naturally.

Prerequisites

A working understanding of the log-normal distribution, parametric survival regression, and the accelerated-failure-time (AFT) parameterisation as an alternative to the proportional-hazards form.

Theory

The log-normal survival function is

$$S(t) = 1 - \Phi\left(\frac{\log t - \mu}{\sigma}\right),$$

with hazard

$$h(t) = \frac{(1/(t\sigma))\phi((\log t - \mu)/\sigma)}{1 - \Phi((\log t - \mu)/\sigma)}.$$

The hazard has a unique maximum near $t \approx \exp(\mu - \sigma^2)$ for $\sigma > 1$ and decreases thereafter. The AFT parameterisation $\log T = X\beta + \sigma W$ (with $W \sim \mathcal{N}(0,1)$) makes regression coefficients interpretable as multiplicative effects on the *event time* — a one-unit increase in a covariate scales the time scale by $\exp(-\beta)$.

Assumptions

A non-monotone (rising-then-falling) hazard, log-Normal residual structure on the log time scale, independent observations, and non-informative censoring.

R Implementation

```r
library(survival); library(flexsurv)

data(lung)
fit <- survreg(Surv(time, status) ~ age + sex, data = lung, dist = "lognormal")
summary(fit)

# flexsurv
flex <- flexsurvreg(Surv(time, status) ~ age + sex, data = lung, dist = "lnorm")
plot(flex, type = "hazard")
```

Output & Results

`survreg()` returns AFT coefficients on the log-time scale plus the scale parameter σ. `flexsurvreg()` produces equivalent estimates with cleaner survival, hazard, and quantile prediction methods. `plot(flex, type = "hazard")` draws the model-implied hazard curve, making the rise-peak-decline shape visible against KM-based hazard estimates.

Interpretation

A reporting sentence: "A log-normal AFT model estimated that each additional year of age shortened expected event time by approximately 2 % ($\hat{\beta}_{\text{age}} = -0.02$, 95 % CI -0.04 to -0.00); the model-implied hazard peaked at approximately 9 months and declined thereafter, consistent with the observed concentration of events in the first year of follow-up." Always plot the implied hazard alongside Kaplan-Meier to verify the non-monotone shape is supported by the data.

Practical Tips

- Use log-normal when Kaplan-Meier or non-parametric hazard estimates show a clear "hump" of events early in follow-up; for monotone hazards, Weibull or exponential are simpler and more efficient.
- Log-logistic is a close alternative with similar non-monotone hazard but heavier tails; compare AICs across both before committing.
- AIC comparison across parametric families (exponential, Weibull, log-normal, log-logistic, generalised gamma) is the standard model-selection workflow; the generalised gamma nests several of these.
- For very flexible hazards beyond the log-normal shape, Royston-Parmar spline-based models offer more freedom while remaining parametric.
- Predictions on the natural time scale come via `predict(fit, type = "quantile")` for `survreg` or `summary(flex, type = "quantile")` for `flexsurv` objects.
- When some covariates produce monotone hazards and others produce non-monotone, consider a heterogeneous-effects extension or a flexible spline model rather than forcing one parametric family.

R Packages Used

`survival` for `survreg()` with `dist = "lognormal"`; `flexsurv` for `flexsurvreg()` with cleaner prediction interfaces and rich plotting; `eha` for additional parametric hazard models; `rstpm2` for Royston-Parmar spline-based parametric extensions when the log-normal hazard shape is too restrictive.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Survival primer: KM, log-rank, Cox PH
- Time-varying covariates and immortal-time bias
- Competing risks and multistate models
- Survival ML (random survival forest, DeepSurv)
- Time-dependent Brier, IPA, external validation